E^2-Bench: Eval of Eval – Measuring Self-Validation Reliability of LLM Agents Beyond Test Passing

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Abstract

As Large Language Models (LLMs) are increasingly deployed as autonomous agents, their ability to self-validate their own outputs has become a critical bottleneck. While existing benchmarks evaluate whether an agent can complete a task, they rarely measure whether the agent *knows* if it has completed the task correctly. Recent studies show that up to 50% of AI-generated code that passes automated tests is still rejected by human reviewers due to quality issues. To address this gap, we introduce **E^2-Bench (Eval of Eval)**, a comprehensive benchmark designed to measure the self-validation reliability of LLM agents across three distinct cognitive levels: (1) **Logical Validation** (correctness of code, math, and data analysis), (2) **Perceptual Validation** (detection of UI/UX and visual quality defects), and (3) **Consistency Validation** (detection of cross-context contradictions in multi-part artifacts). We evaluate frontier models (GPT-4.1-mini and Gemini 2.5 Flash) on 800 tasks. Our results reveal a severe "Yes Man" bias at the logical level (False Positive Rates of 66-100%), an over-reporting tendency at the perceptual level (100% recall but 25-35% precision), and moderate reliability at the consistency level. These findings demonstrate that current LLMs are highly unreliable judges of their own work, highlighting the urgent need for better self-validation mechanisms before agents can be safely deployed in production.

1 Introduction

The paradigm of AI development is shifting from single-turn chat assistants to autonomous agents capable of executing complex, multi-step workflows. A core requirement for true autonomy is *self-validation*—the ability of an agent to inspect its own generated artifacts, identify errors, and determine whether a task has been successfully completed before delivering it to the user.

However, recent empirical evidence suggests a significant gap between an agent's ability to generate a solution and its ability to evaluate it. A March 2026 study by METR found that many pull requests generated by agents that pass all automated tests in SWE-bench would still not be merged by human maintainers due to subtle quality issues, edge cases, or architectural regressions [1]. This highlights a fundamental truth: **passing automated tests does not equal delivery-quality work**.

To systematically quantify this gap, we propose **E^2-Bench (Eval of Eval)**, a novel benchmark that shifts the evaluation focus from task completion to self-validation reliability. Unlike existing benchmarks that measure whether an LLM can write code or solve math problems, E^2-Bench measures whether an LLM can accurately judge if a given solution (often its own) is correct, high-quality, and internally consistent.

We structure E^2-Bench around a three-level taxonomy of validation, mirroring the cognitive checks a human reviewer performs:

1. **Level 1: Logical Validation.** Can the agent verify objective correctness? (e.g., Does the code handle edge cases? Is the math correct?)

2. **Level 2: Perceptual Validation.** Can the agent detect "common sense" quality issues that a human would notice immediately? (e.g., Are UI elements overlapping? Is the text contrast too low?)

3. **Level 3: Consistency Validation.** Can the agent ensure that different parts of a complex artifact do not contradict each other? (e.g., Does the README match the code implementation?)

Our pilot evaluation of GPT-4.1-mini and Gemini 2.5 Flash reveals alarming vulnerabilities. At the logical level, models exhibit an extreme "Yes Man" bias, approving incorrect solutions 66% to 100% of the time. At the perceptual level, models can find planted UI bugs when prompted, but suffer from severe hallucination, generating massive amounts of false alarms (precision $<$ 36%).

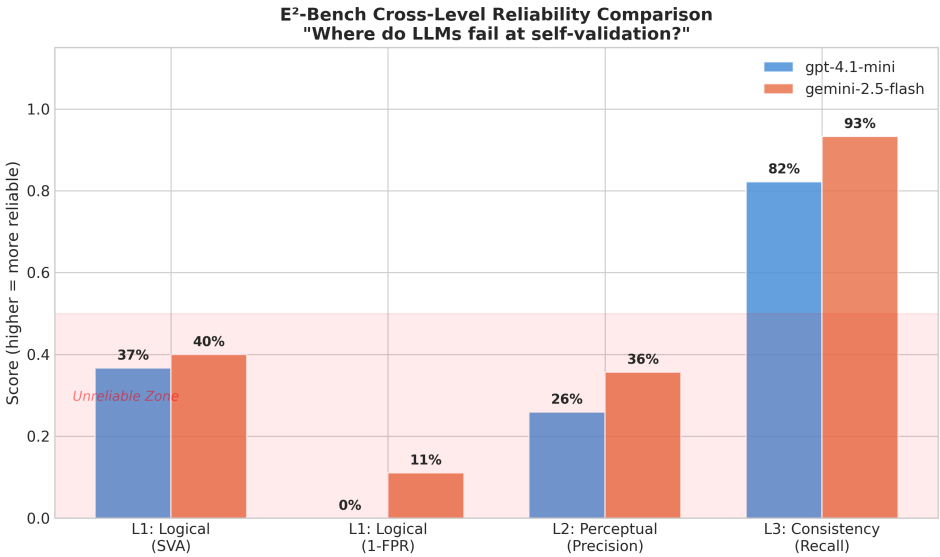

Figure 1: Cross-level reliability comparison. Models struggle significantly with logical self-validation (high False Positive Rate) and perceptual precision, falling into the "Unreliable Zone."

2 The Three-Level Validation Framework

2.1 Level 1: Logical Validation

Logical validation tests the agent's ability to verify objective correctness where a definitive ground truth exists. We evaluate this across three domains: Code Generation, Logical Reasoning (e.g., Game of 24), and Data Analysis.

Methodology: We use a Two-Stage Forced Verification Protocol. In Stage 1, the model generates a solution. In Stage 2, the model is presented with its own solution and asked to rigorously review it, generate mental test cases, and make a binary PASS/FAIL decision. We then compare the model's decision against hidden automated tests or ground truth answers.

2.2 Level 2: Perceptual Validation

Perceptual validation tests the agent's ability to detect quality issues that are functionally "correct" (e.g., valid HTML/CSS) but practically flawed from a human user's perspective.

Methodology: We utilize a *Planted Bug Detection* paradigm. We programmatically inject common UI/UX defects (e.g., stretched images, unreadable text contrast, overlapping z-indexes)

into valid HTML artifacts. The model is provided with the source code and asked to identify visual bugs. We measure Recall (did it find the planted bugs?) and Precision (did it hallucinate non-existent bugs?).

2.3 Level 3: Consistency Validation

Consistency validation tests the agent's ability to maintain a coherent mental model across long contexts and multi-part artifacts.

Methodology: We generate multi-part documents (e.g., a financial report and its accompanying data table, or a codebase and its README). We then inject subtle but critical contradictions between the parts (e.g., the report claims a 15% increase, but the data shows a 12% decrease). The model must cross-reference the parts and identify the contradictions.

3 Experiments and Results

We evaluated GPT-4.1-mini and Gemini 2.5 Flash on a subset of the E^2-Bench dataset (100 tasks across all three levels).

3.1 Level 1 Results: The "Yes Man" Bias

The most striking finding at Level 1 is the extreme False Positive Rate (FPR). When presented with an incorrect solution, GPT-4.1-mini approved it 100% of the time across all three domains. Gemini 2.5 Flash showed slightly more skepticism in code (66.7% FPR) but still failed completely in reasoning and data analysis (100% FPR).

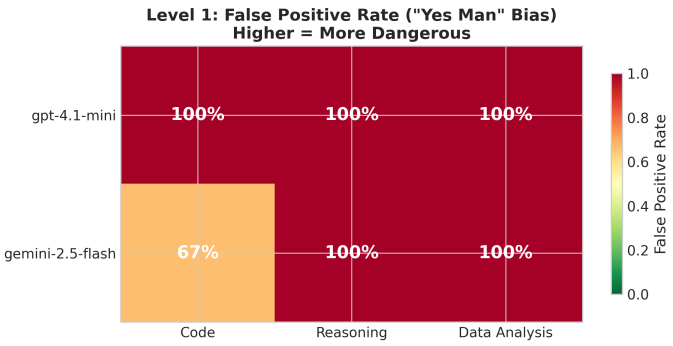

Figure 2: False Positive Rate (FPR) at Level 1. An FPR of 100% means the model *never* correctly identified a failing solution as a failure.

This indicates that current LLMs lack the critical thinking required for self-correction. They act as "Yes Men" to their own outputs, assuming that if they generated it, it must be correct.

3.2 Level 2 Results: The Over-Reporting Problem

At the perceptual level, both models achieved 100% recall, successfully identifying all planted UI bugs (such as image sizing issues and text contrast problems) directly from the HTML source code.

However, this high recall came at the cost of catastrophic precision (25.9% for GPT-4.1-mini, 35.7% for Gemini). The models hallucinated dozens of false alarms, flagging perfectly valid CSS choices as "bugs." This suggests that while LLMs possess the semantic knowledge of what constitutes a UI bug, they lack the grounded visual context to accurately determine if a bug is actually manifesting.

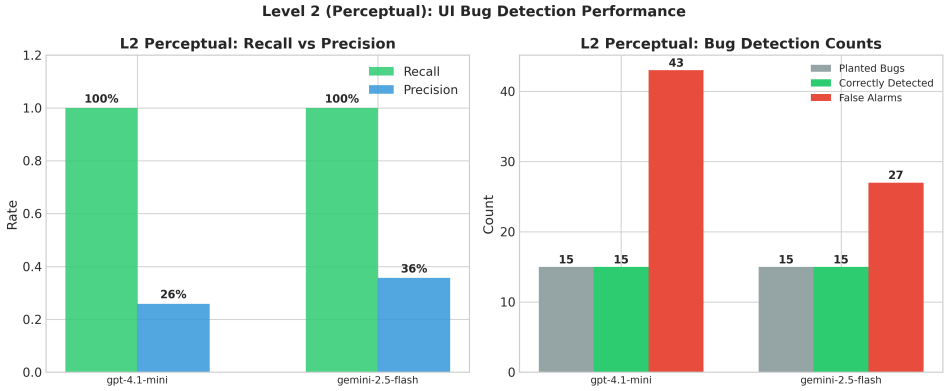

Figure 3: Level 2 Perceptual Validation performance. Both models show perfect recall but terrible precision, generating far more false alarms than actual detections.

3.3 Level 3 Results: Consistency Detection

Models performed best at Level 3. Gemini 2.5 Flash achieved 93.3% recall and 95.5% precision in finding cross-context contradictions, while GPT-4.1-mini achieved 82.2% recall and 97.4% precision. This indicates that LLMs are highly capable of cross-referencing text for logical consistency, making this the most reliable form of self-validation currently available.

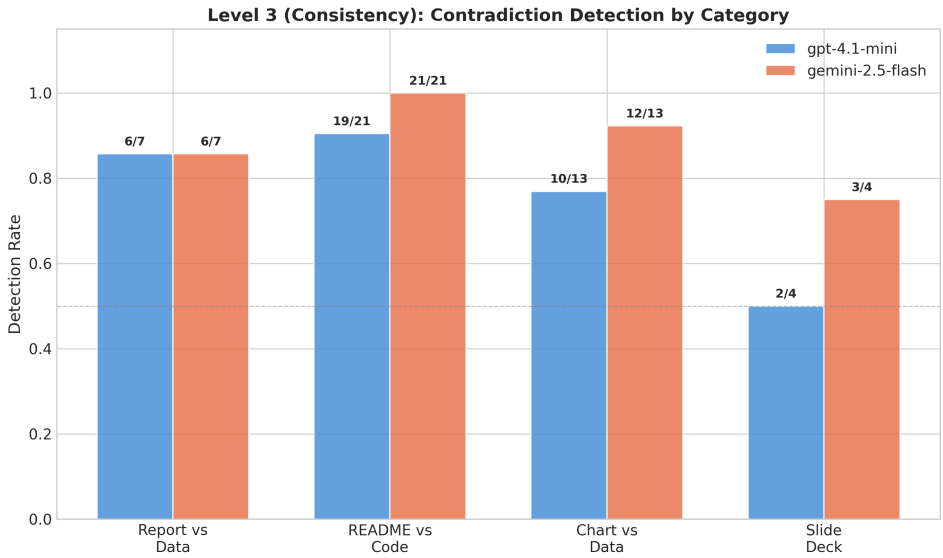

Figure 4: Level 3 Consistency Validation detection rates by artifact category.

4 Conclusion

E^2-Bench demonstrates that LLM agents are currently unreliable judges of their own work. The severe "Yes Man" bias in logical validation and the massive false alarm rate in perceptual validation pose significant risks for autonomous deployment. Our three-level framework provides a comprehensive tool for the community to measure and improve agent self-validation capabilities beyond simple test-passing metrics.

References

[1] METR. (2026). Many SWE-bench passing PRs would not be merged into main. https://metr.org/notes/2026-03-10-many-swe-bench-passing-prs-would-not-be-merged-into-main/